

π **SCHOOL
OF MATH** x **ALGEBRA 2****CLASS 14**

Rational Equations

Solving & Identifying Extraneous Roots

$$\frac{x+1}{x-2}$$

$$x-2$$

**Domain
Restrictions****Solving
Rational Equations****Checking
Every Solution***Build understanding. Solve with confidence.*schoolofmath.us π

1. Understanding Rational Equations

Rational equations — solving and identifying extraneous roots

A **rational equation** contains a ratio of polynomials.
In this lesson, the variable appears in a denominator.

RATIONAL EXPRESSION

$$\frac{3}{x-2}$$

Represents a value.

RATIONAL EQUATION

$$\frac{3}{x-2} = 6$$

States that two expressions are equal.

A FIRST EXAMPLE

$$\frac{3}{x-2} = 6$$

Restriction: $x \neq 2$

$$3 = 6(x-2)$$

$$3 = 6x - 12$$

$$x = \frac{5}{2}$$

Check:

$$3 \div \left(\frac{5}{2} - 2\right) = 6 \quad \checkmark$$



A denominator can never equal zero.
Always check solutions in the original equation.

Image 1 practice

Understanding rational equations

Q1. Which is an expression and which is an equation: $\frac{7}{x+4}$ and $\frac{7}{x+4} = 2$? Give the restriction.

Q2. Can $x = 0$ be tested as a solution of $\frac{x+6}{x} = 3$? Explain before solving.

Q3. Solve $\frac{8}{x} = 2$. State the restrictions and check every candidate.

Q4. Solve $\frac{12}{x+1} = 3$. State the restrictions and check every candidate.

Q5. Solve $\frac{2x+5}{x-1} = 3$. State the restrictions and check every candidate.



2. Identifying Domain Restrictions

Find excluded values before solving.

A **rational expression** is undefined when its denominator equals zero. These values are excluded from the domain.

1

Factor each denominator.

2

Set each factor equal to zero.

3

Exclude those values.

EXAMPLE: TWO DENOMINATORS

$$\frac{2}{x-3} = \frac{x+1}{x+4}$$

$$x - 3 = 0 \rightarrow x = 3$$

$$x + 4 = 0 \rightarrow x = -4$$

Restrictions: $x \neq 3$ and $x \neq -4$

FACTOR TO REVEAL EVERY RESTRICTION

$$\frac{5}{x^2 - 9} = \frac{5}{(x-3)(x+3)}$$

$$x^2 - 9 = (x-3)(x+3)$$

Restrictions: $x \neq 3$ and $x \neq -3$

CANCELLATION DOES NOT REMOVE A RESTRICTION

$$\frac{x-2}{x-2} = 1, \quad x \neq 2$$

Even after simplifying,
 $x = 2$ remains excluded.



Keep the restrictions from the **ORIGINAL** denominators.

Image 2 practice

Identifying domain restrictions

Q6. State the restriction for $\frac{9}{x-6} = 2$. Do not solve.

Q7. Find the excluded values in $\frac{x+1}{x^2-25}$.

Q8. Find every excluded value in $\frac{5}{2x(x+3)}$.

Q9. State all restrictions for $\frac{2}{x^2-x-12} = \frac{7}{x^2-16}$.

Q10. A student cancels $x - 7$ in $\frac{x^2-49}{x-7}$ and claims every real x is allowed. Correct the claim.



3. Finding the LCD

The Least Common Denominator.

The **LCD** is the simplest polynomial multiple of every denominator.

1

Factor every denominator.

2

Identify the distinct factors.

3

Use each factor at its highest required power.

EXAMPLE 1: SHARED FACTORS

$$\frac{1}{2x(x-3)}, \frac{5}{3x^2}, \frac{4}{x-3}$$

Denominators: $2x(x-3)$, $3x^2$, $x-3$

Numbers: LCM of 2 and 3 is 6.

Factor x : use x^2 .

Factor $(x-3)$: use it once.

$$\text{LCD} = 6x^2(x-3)$$

Restrictions: $x \neq 0, 3$

EXAMPLE 2: FACTOR FIRST

Denominators: $x^2 - 9$ and $(x-3)^2$

$$x^2 - 9 = (x-3)(x+3)$$

Use $(x-3)^2$ and $(x+3)$.

$$\text{LCD} = (x-3)^2(x+3)$$

Restrictions: $x \neq 3, -3$



Include every distinct factor at its highest power.
Do not multiply repeated factors unnecessarily.

Image 3 practice

Finding the least common denominator

Q11. Find the least common denominator for $4x$, $6x^2$. State the excluded values.

Q12. Find the least common denominator for $x + 2$, $x(x + 2)$. State the excluded values.

Q13. Find the least common denominator for $x^2 - 16$, $x - 4$. State the excluded values.

Q14. Find the least common denominator for $(x - 1)^2$, $x^2 - 1$. State the excluded values.

Q15. Find the least common denominator for $6x(x + 4)$, $9x^2(x + 4)^2$. State the excluded values.



4. Clearing Denominators

Multiply EVERY term by the LCD.

Multiplying both sides by the LCD removes the fractions.
Keep all original domain restrictions.

1

IDENTIFY THE RESTRICTIONS
AND LCD

$$\frac{2}{x} + \frac{1}{3} = \frac{5}{x}$$

Restriction: $x \neq 0$

$$\text{LCD} = 3x$$

2

MULTIPLY EVERY TERM BY $3x$

$$(3x)\left(\frac{2}{x}\right) + (3x)\left(\frac{1}{3}\right) = (3x)\left(\frac{5}{x}\right)$$

3

SIMPLIFY EACH PRODUCT

$$6 + x = 15$$

The denominators cancel
because $x \neq 0$.

$$x = 9$$

4

CHECK IN THE ORIGINAL EQUATION

$$\frac{2}{9} + \frac{1}{3} = \frac{5}{9}$$

$$\frac{2}{9} + \frac{3}{9} = \frac{5}{9}$$



Valid solution: $x = 9$



Common mistake:

Do not multiply only the fractions containing x .
The term $\frac{1}{3}$ must also be multiplied by $3x$.

Every term. Both sides. Keep the restrictions.

Image 4 practice

Clearing denominators

Q16. Solve $\frac{3}{x} + \frac{1}{2} = \frac{7}{x}$. State the restrictions and check every candidate.

Q17. Solve $\frac{2}{x+1} + \frac{1}{3} = \frac{5}{x+1}$. State the restrictions and check every candidate.

Q18. Solve $\frac{1}{x} + \frac{1}{x+2} = \frac{3}{x(x+2)}$. State the restrictions and check every candidate.

Q19. Solve $\frac{4}{x-3} - \frac{1}{x+3} = \frac{18}{x^2-9}$. State the restrictions and check every candidate.

Q20. Solve $\frac{x+1}{x-2} = 3 + \frac{5}{x-2}$. State the restrictions and check every candidate.



5. Equations That Become Linear

Clear the denominators, then solve for x .

A rational equation can become a linear equation after multiplying every term by the LCD.

1 STATE THE RESTRICTION

$$\frac{3}{x-1} + 2 = \frac{7}{x-1}$$

Restriction: $x - 1 \neq 0$

$$x \neq 1$$

$$\text{LCD} = x - 1$$

2 MULTIPLY EVERY TERM BY THE LCD

$$(x-1) \cdot \frac{3}{x-1} + 2(x-1) = (x-1) \cdot \frac{7}{x-1}$$

$$3 + 2(x-1) = 7$$

3 SOLVE THE LINEAR EQUATION

$$3 + 2x - 2 = 7$$

$$2x + 1 = 7$$

$$2x = 6$$

$$x = 3$$

Distribute before combining like terms.

4 CHECK IN THE ORIGINAL EQUATION

$$\frac{3}{3-1} + 2 = \frac{7}{3-1}$$

$$\frac{3}{2} + 2 = \frac{7}{2}$$



$x = 3$ satisfies the equation and does not violate $x \neq 1$.

Solution: $x = 3$



Key Takeaway:

Restrictions $\rightarrow$ LCD $\rightarrow$ Linear equation $\rightarrow$ Check.

Image 5 practice

Equations that become linear

Q21. Solve $\frac{5}{x-2} = 1$. State the restrictions and check every candidate.

Q22. Solve $\frac{2}{x+3} + 3 = \frac{11}{x+3}$. State the restrictions and check every candidate.

Q23. Solve $\frac{3}{x-4} = \frac{5}{x+2}$. State the restrictions and check every candidate.

Q24. Solve $\frac{x+2}{x-3} = 2 + \frac{8}{x-3}$. State the restrictions and check every candidate.

Q25. Solve $\frac{2x+1}{x+4} = 3 - \frac{8}{x+4}$. State the restrictions and check every candidate.

6. Equations That Become Quadratic

Clear denominators, then solve the quadratic.

After clearing denominators, some rational equations contain an x^2 term.

Consider: $x + \frac{6}{x} = 5$

Restriction: $x \neq 0$
LCD = x

1 CLEAR THE DENOMINATOR

$$x \cdot x + x \cdot \frac{6}{x} = 5x$$

$$x^2 + 6 = 5x$$

2 WRITE IN STANDARD FORM

Move every term to one side.

$$x^2 - 5x + 6 = 0$$

3 FACTOR AND SOLVE

$$(x - 2)(x - 3) = 0$$

Candidates: $x = 2$ or $x = 3$

If factoring is difficult,
use the quadratic formula.

4 CHECK BOTH CANDIDATES

$$x = 2: 2 + \frac{6}{2} = 5 \quad \checkmark$$

$$x = 3: 3 + \frac{6}{3} = 5 \quad \checkmark$$

Both values satisfy the original equation
and neither equals 0.

Solutions: $x = 2$ and $x = 3$



Key Takeaway:

A quadratic gives candidate solutions.
Check each one in the original rational equation.

Image 6 practice

Equations that become quadratic

Q26. Solve $x + \frac{12}{x} = 7$. State the restrictions and check every candidate.

Q27. Solve $x - \frac{20}{x} = 1$. State the restrictions and check every candidate.

Q28. Solve $x + \frac{18}{x} = 9$. State the restrictions and check every candidate.

Q29. Solve $2x + \frac{15}{x} = 11$. State the restrictions and check every candidate.

Q30. Solve $x + \frac{4}{x-2} = 7$. State the restrictions and check every candidate.

7. Identifying Extraneous Roots

A candidate is not a solution until it passes the check.

An extraneous root satisfies a transformed equation but does not satisfy the original equation.

Consider: $\frac{x^2 - 4}{x - 2} = 5$

Restriction: $x \neq 2$
 $x - 2 \neq 0$

1

CLEAR DENOMINATORS AND SOLVE

$$x^2 - 4 = 5(x - 2)$$

$$x^2 - 5x + 6 = 0$$

$$(x - 2)(x - 3) = 0$$

Candidates: $x = 2$ or $x = 3$

2

CHECK $x = 2$

$$\frac{2^2 - 4}{2 - 2} = \frac{0}{0}$$

UNDEFINED

The original denominator is zero.



Reject $x = 2$:
extraneous root.

3

CHECK $x = 3$

$$\frac{3^2 - 4}{3 - 2} = \frac{5}{1} = 5$$



The original equation is true.

Accept $x = 3$:
valid solution.

Solution set: $\{3\}$



What if every candidate is rejected?
Then the original equation has no solution.

Key Takeaway:
Check every candidate in the
ORIGINAL equation.

Image 7 practice

Checking and rejecting extraneous roots

Q31. Solve $\frac{x^2-9}{x-3} = 8$. State the restrictions and check every candidate.

Q32. Solve $\frac{x^2-16}{x-4} = 8$. State the restrictions and check every candidate.

Q33. Solve $\frac{x^2+x-12}{x+4} = 2$. State the restrictions and check every candidate.

Q34. Solve $\frac{x^2-1}{x-1} = 3x - 1$. State the restrictions and check every candidate.

Q35. Solve $\frac{x^2-25}{x^2-3x-10} = 2$. State the restrictions and check every candidate.

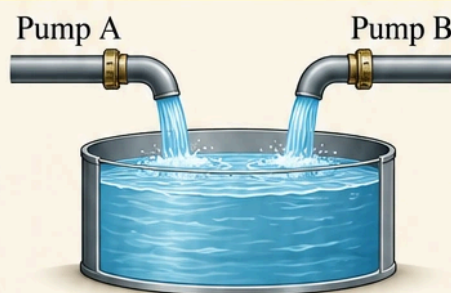
8. Rational Equations in Word Problems

Turn a real situation into an equation.

Filling a water tank

Pump A fills a tank in 6 hours. Together, pumps A and B fill it in 4 hours. How long would pump B take alone?

Assume constant rates and an initially empty tank.


1

DEFINE THE UNKNOWN AND RATES

Let x be the hours pump B takes alone; $x > 0$.

Pump A:
 $\frac{1}{6}$ tank per hour

Pump B:
 $\frac{1}{x}$ tank per hour

Together:
 $\frac{1}{4}$ tank per hour

2

BUILD THE EQUATION

$$\frac{1}{6} + \frac{1}{x} = \frac{1}{4}$$

Individual rates add to the combined rate.

3

CLEAR DENOMINATORS AND SOLVE

$$\text{LCD} = 12x$$

$$12x \left(\frac{1}{6} \right) + 12x \left(\frac{1}{x} \right) = 12x \left(\frac{1}{4} \right)$$

$$2x + 12 = 3x$$

$$x = 12$$

4

CHECK AND INTERPRET

$$\frac{1}{6} + \frac{1}{12} = \frac{2}{12} + \frac{1}{12} = \frac{1}{4}$$



Pump B takes 12 hours to fill the tank alone.

Together, the pumps work faster than either pump alone.



Key Takeaway: Add rates, not completion times.

Other applications: travel time = distance $\div$ speed; shared cost = total cost $\div$ people.

Image 8 practice

Rational equations in word problems

Q36. Printer A produces one batch in 8 minutes. Two printers working together produce that batch in 3 minutes. How long would printer B take alone?

Q37. One machine completes a job in 10 hours and another in 15 hours. How long do they take together?

Q38. A \$180 coach fee is shared equally. If 3 more students join, each pays \$5 less. How many students were in the original group?

Q39. A cyclist travels 60 km at a steady speed. Increasing that speed by 5 km/h would reduce the time by 1 hour. Find the original speed.

Q40. A slower pump takes 3 hours longer than a faster pump to fill a tank. Together they fill it in 2 hours. Find each pump's time alone.



9. Proportions & Cross-Multiplication

Use this shortcut when each side is a single fraction.

Cross-Multiplication Rule

$$\frac{a}{b} = \frac{c}{d} \text{ implies } ad = bc$$

Valid when $b \neq 0$ and $d \neq 0$.

The products are:

- $a \cdot d$ (numerator of left times denominator of right)
- $b \cdot c$ (denominator of left times numerator of right)

Example

$$\frac{3}{x-1} = \frac{2}{x+2}$$

1

STATE THE RESTRICTIONS

Denominators cannot be zero:

$$x \neq 1 \quad \text{and} \quad x \neq -2$$

2

CROSS-MULTIPLY

$$3(x+2) = 2(x-1)$$

Multiply each numerator by the opposite denominator.

3

SOLVE

$$3x + 6 = 2x - 2$$

$$x + 6 = -2$$

$$x = -8$$

Solution: $x = -8$

4

CHECK IN THE ORIGINAL EQUATION

Left: $\frac{3}{-8-1} = \frac{3}{-9} = -\frac{1}{3}$

Right: $\frac{2}{-8+2} = \frac{2}{-6} = -\frac{1}{3}$



Both sides match, and -8 is allowed.



Before you cross-multiply:

If either side contains a sum of fractions, first combine it into one fraction or multiply every term by the LCD.



Key Takeaway: One fraction on each side. Restrictions first. Check last.

Image 9 practice

Proportions and cross-multiplication

Q41. Solve $\frac{4}{x} = \frac{2}{7}$. State the restrictions and check every candidate.

Q42. Solve $\frac{5}{x-2} = \frac{3}{x+4}$. State the restrictions and check every candidate.

Q43. Solve $\frac{x+2}{5} = \frac{x-1}{3}$. State the restrictions and check every candidate.

Q44. Solve $\frac{x+1}{x-2} = \frac{x+5}{x+2}$. State the restrictions and check every candidate.

Q45. Solve $\frac{x}{x-1} = \frac{6}{x+1}$. State the restrictions and check every candidate.

10. Infinitely Many Solutions

An identity is true for every allowed value.

Some rational equations simplify to an identity.
Their solutions include every value in the original domain.

Example

$$\frac{x+1}{x-2} = 1 + \frac{3}{x-2}$$

1 STATE THE RESTRICTION

The denominator cannot be zero:

$$x \neq 2$$

$$\text{LCD} = x - 2$$

2 CLEAR THE DENOMINATORS

$$x + 1 = (x - 2) + 3$$

$$x + 1 = x + 1$$

$$0 = 0$$

3 INTERPRET THE RESULT

The final statement is always true.

**Infinitely many solutions:
all real x except 2.**

Solution set: $(-\infty, 2) \cup (2, \infty)$

4 CHECK AN ALLOWED VALUE

For $x = 3$:

$$\frac{3+1}{3-2} = 1 + \frac{3}{3-2}$$

$$4 = 4 \checkmark$$

This illustrates the identity; the algebra proves it for every $x \neq 2$.

RECOGNIZE THE THREE OUTCOMES

Valid candidates after checking



one or more specific solutions

A contradiction such as $0 = 5$



no solution

An identity such as $0 = 0$



all allowed values



An identity does NOT restore excluded values.
 $x = 2$ is still undefined.

Image 10 practice

Infinitely many solutions and no solution

Q46. Solve $\frac{x+4}{x+1} = 1 + \frac{3}{x+1}$. State the restrictions and check every candidate.

Q47. Solve $\frac{2x-6}{x-3} = 2$. State the restrictions and check every candidate.

Q48. Solve $\frac{x+5}{x-2} = 1 + \frac{6}{x-2}$. State the restrictions and check every candidate.

Q49. Solve $\frac{x^2-9}{x-3} = x + 3$. State the restrictions and check every candidate.

Q50. Solve $\frac{1}{x-1} - \frac{1}{x+1} = \frac{2}{x^2-1}$. State the restrictions and check every candidate.

Homework review

R1–R5

R1. Solve $\frac{15}{x+2} = 3$. State the restrictions and check every candidate.

R2. List every restriction in $\frac{1}{x^2-2x-15} = \frac{4}{x^2-9}$. Do not solve.

R3. Find the least common denominator for $4x(x-2)$, $6x^2(x-2)^2$. State the excluded values.

R4. Solve $\frac{2}{x} + \frac{1}{4} = \frac{5}{x}$. State the restrictions and check every candidate.

R5. Solve $\frac{4}{x-5} = \frac{3}{x+1}$. State the restrictions and check every candidate.

Homework review

R6–R10

R6. Solve $x + \frac{24}{x} = 10$. State the restrictions and check every candidate.

R7. Solve $\frac{x^2-36}{x-6} = 10$. State the restrictions and check every candidate.

R8. One machine completes a batch in 12 minutes. Together with a second machine it completes a batch in 5 minutes. Find the second machine's time alone.

R9. Solve $\frac{x+2}{x-1} = \frac{x+6}{x+1}$. State the restrictions and check every candidate.

R10. Solve $\frac{x+7}{x+2} = 1 + \frac{5}{x+2}$. State the restrictions and check every candidate.

Homework word problems

W1–W5

W1. A group shares a \$96 museum fee equally. Each person pays \$12. How many people are in the group?

W2. A hiker covers 18 miles at a constant speed and takes 6 hours. What is the hiking speed?

W3. A club's \$150 transport cost is shared equally. Each student also pays a \$5 ticket fee, for a total of \$20 each. How many students are attending?

W4. Pump A fills a tank in 9 hours. A second pump working with A reduces the filling time to 6 hours. How long does the second pump take alone?

W5. A workshop model defines usable material per panel as $A(x) = \frac{x^2-16}{x-4}$, where x is a dimension in meters and $x > 0$. Which x makes $A(x) = 10$ meters? Identify any extraneous root from clearing the denominator.

Homework word problems

W6–W10

W6. A coach costs \$240 to hire. If 4 extra students join, each student's share decreases by \$5. How many students were in the original group?

W7. A train travels 120 km at a steady speed. At a speed 10 km/h faster, it would arrive 1 hour sooner. Find its original speed.

W8. One machine needs 5 hours longer than another to complete a job. Together they finish in 6 hours. How long does each take alone?

W9. A boat travels 24 km downstream and 24 km upstream in 5 hours total. The current is 2 km/h. Find the boat's speed in still water, assuming constant speeds.

W10. Two pumps fill a tank together in 6 hours. Working alone, the slower pump takes 7 hours longer than the faster pump. Find both individual times.